A computer-assisted determination of $C(12, 6, 4)$

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Abstract

Let $C(v, k, t)$ be the minimum size of a family of k-subsets of a v-set that contains every t-subset. We prove, with independently checkable SAT certificates, that

$$C(12, 6, 4) = 41.$$

The published upper bound is a 41-block cover. For the lower bound, a hypothetical 40-block cover is forced into a rigid equality case. Its six pair-degree excesses form a perfect matching and every triple has degree three or four. Fixing one nonmatching pair, the nine blocks through it form an optimal $(10, 4, 2)$ covering in which two specified points have degree four. There are 119,700 such labelled links, forming 266 orbits under the order-768 stabilizer of the matching and the fixed pair. A deterministic enumerator and a structurally different independent auditor establish this partition. Each canonical link gives one SAT instance; all 266 are unsatisfiable, with every LRAT proof independently replayed by two checkers. The only auxiliary mathematical ingredient is the small local theorem $C(10, 4, 2) = 9$, together with the fact that every point in a nine-block optimum has degree three or four. Its sole finite case is supplied both as a short exhaustive program and as a replayed LRAT certificate.

1. Definitions and certificate convention

A (v, k, t) covering is a family $\mathcal{B}$ of k-subsets ("blocks") of a v-element point set such that every t-subset lies in at least one block. Repeated blocks can be discarded, so blocks may be assumed distinct. The covering number $C(v, k, t)$ is the least possible $|\mathcal{B}|$.

For a subset S of the point set, write

$$r_S = |\{B \in \mathcal{B} : S \subseteq B\}|.$$

For a point x, write r_x; for a pair xy, write $\lambda_{xy} = r_{\{x,y\}}$.

The finite unsatisfiability claims below are not accepted merely because a solver reports `UNSAT`. For each claim:

1. a deterministic generator writes a pinned DIMACS CNF;
2. a separate fail-closed auditor reconstructs its mathematical objects and its cardinality clauses;
3. CaDiCaL writes a DRAT proof;
4. `drat-trim` checks the DRAT proof and exports LRAT; and
5. the independent `lrat-check` program checks that LRAT file against the pinned CNF; and
6. the CakeML-extracted `cake_lpr` checker replays the same LRAT bytes.

Every source file, instance, result log, and distributable certificate used for a finite claim is covered by a SHA-256 manifest. The manifests are tied to the published Git commit and release; as usual, a manifest does not hash itself.

2. The local theorem

Lemma 1

$$C(10,4,2) = 9.$$

Moreover, in every nine-block $(10,4,2)$ covering, every point has degree three or four. Consequently its degree multiset is $4^6, 3^4$.

Proof

First suppose eight blocks cover all pairs. A block through a point covers only three of the nine pairs incident with that point, so every point has degree at least three. Since the degree sum is $8 \cdot 4 = 32$, the degree sequence is either

$$5, 3^9 \qquad \text{or} \qquad 4^2, 3^8.$$

If $r_x = 3$, then

$$\sum_{y \neq x} \lambda_{xy} = 3r_x = 9.$$

All nine terms are positive, hence every one is equal to one.

In the first degree pattern, let a be the degree-five point. Every other point has degree three, so $\lambda_{ay} = 1$ for all nine $y \neq a$. This gives $\sum_{y \neq a} \lambda_{ay} = 9$, contradicting $3r_a = 15$.

In the second pattern, let a, b be the degree-four points and let L be the other eight points. Then

$$12 = 3r_a = \lambda_{ab} + \sum_{y \in L} \lambda_{ay} = \lambda_{ab} + 8,$$

so $\lambda_{ab} = 4$. Thus all four blocks through a contain b; after deleting a, b, their four pairs partition L. The remaining four blocks lie in L, and every point of L occurs twice in them. The four already covered pairs leave 24 uncovered pairs in L, exactly the total pair capacity of four 4-subsets. Hence those four blocks may not repeat a pair, so any two of them intersect in at most one point. On the other hand, counting their pairwise intersections by points gives

$$\sum_{1 \leq i < j \leq 4} |B_i \cap B_j| = \sum_{y \in L} \binom{2}{2} = 8,$$

whereas six pairs of blocks with intersections at most one give a total at most six. This contradiction excludes eight blocks, and therefore every smaller cover as well.

The following nine blocks give the matching upper bound:

```
0 1 5 6
2 3 4 6
6 7 8 9
1 2 3 7
0 4 5 7
0 2 5 8
0 3 5 9
1 2 4 9
1 3 4 8
```

A direct check covers all 45 pairs. Thus $C(10,4,2) = 9$.

It remains to bound degrees in a nine-block optimum. Suppose a point x has degree r. Delete x. Its r incident blocks become triples on a nine-point set Y, and the remaining $9 - r$ blocks are quadruples. The triples leave at least

$$\binom{9}{2} - 3r = 36 - 3r$$

uncovered edges, while the quadruples have total edge capacity $6(9 - r)$. For $r = 7, 8, 9$, this would respectively require

$$15 \le 12, \qquad 12 \le 6, \qquad 9 \le 0,$$

all impossible.

Let $r = 6$. Equality is forced in the preceding capacity count: the six triples have disjoint edge sets, and the three quadruples partition the remaining 18 edges. For $y \in Y$, let n_y and m_y count incident triples and quadruples. Then

$$8 - 2n_y = 3m_y.$$

Every $y \in Y$ lies in an incident triple, because the original pair $\{x, y\}$ must be covered. Hence $n_y \ge 1$, and the only possibilities are $(n_y, m_y) = (1, 2)$ and $(4, 0)$. Since $\sum_y n_y = 18$, exactly three points are of the latter type. Call them heavy. If k_i is the number of heavy points in the i-th triple, then $\sum_i k_i = 12$. Disjointness of triple edges gives

$$\sum_i \binom{k_i}{2} \le \binom{3}{2} = 3,$$

but $\binom{k}{2} \ge k - 1$ for $0 \le k \le 3$, and therefore

$$\sum_i \binom{k_i}{2} \ge \sum_i (k_i - 1) = 12 - 6 = 6.$$

This contradiction excludes degree six.

For $r = 5$, deleting x leaves five distinct triples whose union is Y, together with four quadruples that must cover every edge of K_9 not already covered by the triples. By S_9-symmetry, fix one triple as $\{0, 1, 2\}$. The verifier enumerates all

$$\binom{83}{4} = 1,837,620$$

choices of the other four triples. For each of the 411,300 choices whose vertex union is Y, a complete recursive search tests whether at most four quadruples cover every residual edge. It deliberately permits repeated quadruples, so it searches a superset of the valid completions. No completion exists. The same finite claim is encoded independently as a CNF and certified by replayed DRAT and LRAT proofs; the precise certificate record is in Section 5.

Thus every point has degree at most four. Every point has degree at least three, and

$$36 = \sum_x r_x = 4n_4 + 3(10 - n_4)$$

gives $n_4 = 6$. This proves the lemma. □

3. The equality case for 40 blocks

The Schönheim inequalities and Lemma 1 give

$$C(11,5,3) \geq \left\lceil \frac{11}{5} C(10,4,2) \right\rceil = 20$$

so the value 20 used here is a direct consequence of Lemma 1, not an external assumption. Consequently,

$$C(12,6,4) \geq \left\lceil \frac{12}{6} C(11,5,3) \right\rceil \geq 40.$$

Suppose for contradiction that a 40-block $(12,6,4)$ cover $\mathcal{B}$ exists. It has no repeated block: deleting one copy of a repeated block would leave a cover with at most 39 blocks, contradicting the lower bound just proved. More generally, any cover with fewer than 40 distinct blocks could be padded with unused 6-subsets to obtain a cover with 40 distinct blocks. Thus excluding the latter excludes every smaller or multiset cover as well.

3.1 Point and pair degrees

The blocks through a pair, after deleting that pair, form a $(10,4,2)$ cover. Because the original blocks are distinct, their pair-deleted 4-subsets are distinct as well. Lemma 1 gives $\lambda_{xy} \geq 9$ for every pair.

At a fixed point x,

$$5r_x = \sum_{y \neq x} \lambda_{xy} \geq 11 \cdot 9 = 99.$$

Integrality gives $r_x \geq 20$. Since

$$\sum_x r_x = 40 \cdot 6 = 240 = 12 \cdot 20,$$

every point has degree exactly 20.

The total pair incidence is

$$\sum_{\{x,y\}} \lambda_{xy} = 40\binom{6}{2} = 600,$$

only six above the baseline $66 \cdot 9 = 594$. At a fixed point x,

$$\sum_{y \neq x} \lambda_{xy} = 5r_x = 100,$$

only one above the local baseline $11 \cdot 9 = 99$. Therefore the nonnegative integral excesses $\lambda_{xy} - 9$ form a weighted graph of degree one at every vertex. They are exactly the six edges of a perfect matching M, each with excess one. Consequently,

$$\lambda_{xy} = \begin{cases} 10, & xy \in M, \\ 9, & xy \notin M. \end{cases}$$

Relabel the points so that

$$M = \{\{0,1\},\{2,3\},\{4,5\},\{6,7\},\{8,9\},\{10,11\}\}.$$

3.2 Triple degrees

Every triple T has degree at least three. Indeed, the nine quadruples $T \cup \{z\}$ must all be covered, while one 6-block containing T covers only three choices of z.

Every triple has degree at most four. Among its three pairs, at least one is not in the matching M. Choose such a pair $P \subset T$. Exactly nine blocks contain P. Deleting P from them produces nine distinct 4-subsets and hence a nine-block $(10, 4, 2)$ covering: every pair Q among the other ten points is covered because the original quadruple $P \cup Q$ is covered. The third point of T has degree r_T in this local cover. Lemma 1 therefore gives $r_T \in \{3, 4\}$.

The degree-four triples satisfy a useful exact identity. For any pair P,

$$\sum_{z \notin P} r_{P \cup \{z\}} = 4\lambda_P,$$

because each block through P supplies four choices of its third point. If $P \in M$, the right side is 40, and all ten summands are at most four. Thus every triple containing a matching pair has degree four. These 60 triples are distinct. If $P \notin M$, the right side is 36, and the ten summands are each three or four. Hence exactly six of them have degree four.

Globally, if N_4 is the number of degree-four triples, then

$$800 = 40\binom{6}{3} = \sum_T r_T = 3\binom{12}{3} + N_4,$$

so $N_4 = 140$. Therefore exactly 80 of the 160 triples containing no matching pair have degree four.

3.3 A distinguished pair link

Fix the nonmatching pair

$$P = \{0, 2\}.$$

Exactly nine blocks contain P. Delete P from each of them. The result is a family

$$\mathcal{L}_P \subseteq \binom{V \setminus P}{4}$$

of nine distinct 4-subsets. It covers every pair of the remaining ten points: for $Q \in \binom{V \setminus P}{2}$, any block covering the quadruple $P \cup Q$ contains P, and its image in $\mathcal{L}_P$ contains Q. Thus $\mathcal{L}_P$ is an optimal $(10, 4, 2)$ covering.

The degree of $y \in V \setminus P$ in this link is the global triple degree $r_{P \cup \{y\}}$. In particular, points 1 and 3 have link degree four, because $\{0, 1, 2\}$ and $\{0, 2, 3\}$ each contain a matching edge. Lemma 1 says that the link-degree multiset is $4^6, 3^4$. Consequently every hypothetical 40-cover determines a nine-block $(10, 4, 2)$ link in which the two specified partner points 1 and 3 both belong to the six-point degree-four set.

4. The global finite exclusion

4.1 The shared Boolean formula

For each 6-subset B of the 12 points, introduce a Boolean variable x_B. The shared formula asserts:

$$\sum_{B} x_B = 40;$$

$$\sum_{B \supseteq P} x_B \le \begin{cases} 10, & P \in M, \\ 9, & P \notin M \end{cases} \quad \text{for every pair } P;$$

$$\sum_{B \supseteq T} x_B \le 4 \quad \text{for every triple } T;$$

and

$$\sum_{B \supseteq Q} x_B \ge 1 \quad \text{for every quadruple } Q.$$

The pair ceilings lose no equality information. Their right-hand sides sum to

$$6 \cdot 10 + 60 \cdot 9 = 600,$$

while any 40 selected blocks have exactly $40\binom{6}{2} = 600$ pair incidences. Thus every ceiling is attained.

Every hypothetical 40-cover therefore gives a satisfying assignment of this shared formula.

4.2 The audited compact cardinality encoding

The exhaustive campaign uses the Python-SAT k-modulo-totalizer at pinned commit `7abb1d6fdafe1d75fe3e5cdfb48f7e07a8a76efd`. The proof package does not treat that implementation as an unexamined oracle. A standard-library-only Python transcription independently reconstructs all auxiliary identifiers and all clauses of the production base byte for byte, without importing Python-SAT or a SAT solver.

For a unary vector Y, let $\pi(Y)$ be the length of its true prefix. At a modulo node of radix p, with upper and lower vectors U, L, decode

$$D(U, L) = p\,\pi(U) + \pi(L).$$

Induction over the totalizer tree proves that every satisfying auxiliary assignment has decoded value at least the number of true input literals; an overflow past a truncated upper vector makes the clauses inconsistent. The root comparator forces $D(U, L)$ at most the requested bound. Conversely, when the

bound holds, assigning every node the exact quotient and remainder of its subtree count satisfies every merge and comparator clause. Hence the existential projection of an `AtMost` encoding is exactly the intended inequality. Complementing the inputs proves `AtLeast`, and the pinned wrapper uses disjoint consecutive auxiliaries for the two halves of equality.

The independent auditor reconstructs the production sections for exactly 40 blocks, the 66 pair ceilings, the 220 triple ceilings, and the 495 quadruple coverage clauses. It obtains 122,519 variables and 356,407 clauses, with full-file SHA-256 `637d9326388f77f933f3d066cd762d8c0ac4b2df0d2260d688e031d55959e7f0`. It also performs 5,376 small exhaustive primary-assignment checks with its own DPLL routine. Separately, a logic-free C++ adapter compiled against the pinned official headers agrees with the transcription on 115 vectors containing 123,220 clauses, including the complete equality section.

As an independent encoding cross-check, the repository also constructs the same shared primary formula using full-equivalence prefix gates

$$s_{i,j} \longleftrightarrow s_{i-1,j} \vee (s_{i-1,j-1} \wedge x_i).$$

A separate auditor truth-table-checks every gate and reconstructs the two coarse symmetry roots. This functional base has 271,068 variables and 1,044,768 clauses; its clause-body SHA-256 is `fc7e5de081b3805cd8c77f27c581b6fd6ab3a97b92a2928c3d60fafd2fb76e73`. These functional artifacts are an encoding-level corroborating check; the exhaustive lower bound uses the compact fixed-link roots and their receipts.

4.3 Exhausting the distinguished links

Let $\Gamma = \mathrm{Stab}(M, P)$, where $P = \{0, 2\}$ is fixed as an unordered pair. An element of Γ may swap the two matching pairs $\{0, 1\}$ and $\{2, 3\}$, may arbitrarily permute the other four matching pairs, and may independently swap the endpoints of those four pairs. Hence

$$|\Gamma| = 2 \cdot 4! \cdot 2^4 = 768.$$

This group acts on the ten points outside P, preserving the condition that points 1 and 3 have link degree four.

The production enumerator lists every nine-block $(10, 4, 2)$ covering and then takes Γ-orbits of the admissible ones. A separately written auditor reconstructs the same finite set in two structurally different ways. First, a canonical minimum-uncovered-pair search over the 210 possible 4-subsets finds exactly 359,100 labelled nine-block covers.

Second, fix the six degree-four points H; the other four points have degree three. Choose one degree-three point as an anchor. Its three incident blocks, after deleting the anchor, partition the other nine points into three triples, giving 280 possibilities. For each of the other three degree-three points, the remaining six points admit ten unordered splits into two triples. The auditor tests all 280,000 option tuples, checks consistency, and supplies all compatible heavy-only blocks. It obtains exactly 1,710 distinct covers for each fixed H. Relabelling over the $\binom{10}{6} = 210$ heavy sets produces

$$1,710 \binom{10}{6} = 359,100$$

distinct covers, exactly the same set as the canonical search. The two specified partner points are heavy in

$$1,710 \binom{8}{4} = 119,700$$

of them.

The auditor constructs the order-768 closure of explicit generators and partitions these 119,700 links by generator traversal. It then cross-checks every resulting orbit against all 768 group images. The result is 266 disjoint orbits whose sizes sum to 119,700. The orbit-size histogram is

orbit size	12	16	24	48	96	128	192	256	384	768
number	1	1	5	12	30	2	60	1	37	117

Equivalently, the stabilizer orders are

stabilizer order	1	2	3	4	6	8	16	32	48	64
number	117	37	1	60	2	30	12	5	1	1

The canonical representative list has SHA-256 `369a844c4590d435fc24f82424a0775d51ce326d243aaf740d025c52c81d652b`. The independent enumerator and auditor produce the same bytes.

4.4 The 266 certified roots

For a canonical link $\mathcal{L}$, append one unit clause for each of the 210 possible blocks through P: nine positive units select the blocks $P \cup L$ with $L \in \mathcal{L}$, and the other 201 units are negative. Thus each root has the same 122,519 variables and 356,617 clauses. The independent auditor reconstructs all 210 units directly from each link, checks that the nine positive blocks are distinct and cover all 45 pairs in the link, and binds the resulting clause bytes to the audited shared base.

Every hypothetical 40-cover gives an admissible link by Section 3.3. Some element of Γ maps that link to one of the 266 canonical representatives. Applying the same relabelling to the whole cover preserves the shared formula and P, so the relabelled cover would satisfy the corresponding root. The 266 roots are therefore exhaustive.

For every root, CaDiCaL produces a DRAT proof. `drat-trim` replays it and exports LRAT. Both the independent C `lrat-check` implementation and the CakeML-extracted `cake_lpr` checker then accept the same LRAT bytes against the pinned CNF. A fail-closed aggregate auditor requires the exact case set $\{0, \ldots, 265\}$, recomputes each CNF from the shared base and its 210 units, verifies all file identities, and rejects a missing, duplicated, misindexed, or hash-mismatched receipt.

All 266 roots are UNSAT. Therefore no 40-block $(12, 6, 4)$ cover exists.

5. Certificate record

No digest in this table is populated from an unverified or partial proof.

Claim	Variables	Clauses	Pinned identity	Checked proof
Local degree-five exclusion	6,816	24,835	`bcb0b4138a34e0fc8b795f24d18d97e98f9d6a8e9f5b90c26e51f9e548a5ba32`	DRAT, C LRAT, and CakeML LRAT verified

Claim	Variables	Clauses	Pinned identity	Checked proof
Compact shared global base	122,519	356,407	`637d9326388f77f933f3d066cd762d8c0ac4b2df0d2260d688e031d55959e7f0`	semantic and byte audit verified
Functional encoding cross-check	271,068	1,044,768	`fc7e5de081b3805cd8c77f27c581b6fd6ab3a97b92a2928c3d60fafd2fb76e73`	gate audit verified
Fixed-link classification	—	—	`369a844c4590d435fc24f82424a0775d51ce326d243aaf740d025c52c81d652b`	independent 359,100/119,700/266 partition audit verified
266 fixed-link roots	122,519 each	356,617 each	`8735ea4aedb5d30fc1fa65ff289f4357bec14d354a993270a620254d30ff575a`	DRAT, C LRAT, and CakeML LRAT verified for every root

The functional encoding is not substituted for any compact fixed-link root. The fixed-link proof archives have SHA-256 identities `67acd4729329ad4656b5a94e8a472b11a14a5c81fccaa468448f12957663af4a` (DRAT) and `c45ea9e621db38f4e25f290cb664352c0b80f1839ad475b422098b12fbbbd615` (LRAT). Their deterministic member index and the aggregate above are bound by the release-assets logical identity `499c345b3d48666b67b17df2dda79b6f7af5e7fa7c05668900eb9cd7b5bee2f6`.

6. The 41-block upper bound

The La Jolla Covering Repository publishes a 41-block $(12, 6, 4)$ cover. The exact block list is included in `data/cover_41.txt`. The independent verifier checks that:

- there are 41 distinct blocks of size six on 12 points;
- every one of the $\binom{12}{4} = 495$ quadruples is covered; and
- the multiplicity histograms agree with the pinned verification record.

Thus $C(12, 6, 4) \leq 41$. Section 4 gives $C(12, 6, 4) \geq 41$, and therefore

$$\boxed{C(12, 6, 4) = 41}.$$

7. Scope and review

This is a computer-assisted proof, not a claim that the result has already passed journal peer review. The theorem depends only on the human reductions, the small auditable generators, and the replayed proof certificates. CP-SAT, Gurobi, and exploratory searches in the evidence directory are corroborating checks and are not used as proof.

The accompanying search record documents the authoritative repository state, indexed-web and GitHub searches, and a direct seven-day X API search. That record supports a novelty check; it cannot prove universal absence from unindexed pages or from X's unavailable full archive.

8. References

1. J. Schönheim, "On coverings," *Pacific Journal of Mathematics* **14** (1964), 1405-1411. doi:10.2140/pjm.1964.14.1405.
2. D. Gordon, *La Jolla Covering Repository*, entry for $C(12, 6, 4)$, accessed 24 July 2026. cover entry.
3. D. Gordon, *La Jolla Coverings Repository*, dataset version 1.2, 24 April 2026. doi:10.5281/zenodo.19735294.

9. Computational disclosure

Dennison Bertram directed and published this work. OpenAI Codex assisted with problem selection, proof exploration, implementation, certificate orchestration, auditing, and drafting. Claude Fable was used as an adversarial reviewer at xhigh effort. Claude Opus 5 was separately used at xhigh effort for an adversarial certificate audit and an independent search for new mathematical structure. No language-model assertion is used as proof: the finite claims are represented by deterministic source, pinned instances, hash manifests, and independently replayable certificates. The complete review record and tool provenance accompany this note.